

RAGHAV KACHROO

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Software engineer drawn to making ML systems faster in production - coming from distributed systems and data pipelines, working down the stack toward inference optimization, compilers, and GPUs.

EXPERIENCE

Hao AI Lab, UC San Diego

Jan 2026 – Present

Contributor

San Diego, CA

*[FastVideo](#) — open-source video-generation inference and post-training framework; **4k+** stars, supported as an NVIDIA*

*Dynamo custom worker and forked into SGLang's multimodal-generation stack. **15 merged PRs***

- Traced a “slow” decode stage to a mis-attributed deferred GPU-to-CPU transfer; quantizing frames to uint8 on-device cut end-to-end latency **15.6%** on Cosmos 2.5 2B/A100 at MS-SSIM 1.0.
- Designed **EasyCache**, a model-agnostic residual-skip cache learning a sensitivity factor at runtime and recomputing on accumulated error: **23%** lower Wan2.1 wall time at SSIM **0.946**.
- Split a one-use bootstrap and a resident continuation worker into separate residency, compile, and prewarm policies, cutting a Cosmos 2.5 DFD serving profile **26.2%** (144 → 106s) at PSNR 48.2 dB.
- Registered FlashAttention-2 as a `torch.library` custom op with `register_autograd` across default and varlen paths, making attention traceable under `torch.compile`.
- Decoupled VAE decode precision from encode after bf16 encode regressed causal models, recovering **1.2-1.3x** decode at MS-SSIM 0.9999.

Amazon

Jun 2025 – Sep 2025

Software Development Engineer Intern

Bellevue, WA

- Built a distributed log indexing and query service over **42M+** log entries/hour, running binary-search reads across concurrent Lambda invocations to cut incident triage **15+ min to under 45s**.
- Automated on-call SOPs into AWS Step Functions and Lambda workflows that detect and remediate unattended, saving **12+ engineer-hours per week**.
- Exposed the query service over MCP with a DynamoDB result cache keyed on the normalized query, short-circuiting repeat Lambda calls during concurrent incident investigations.

Aark Global

Apr 2023 – Sep 2024

Software Developer, AI/ML

Delhi, India

- Built an async document ingestion pipeline processing **18,000+ pages/day** via an Azure Queue Storage worker pool pulling files as they landed in a watched directory.
- Implemented a phased Cosmos DB-to-MongoDB migration using dual writes and historical backfill while serving reads from Cosmos DB, maintaining **sub-100ms P95** latency through cutover.
- Built a full-text search pipeline ingesting scanned PDFs through OCR into indexed Elasticsearch documents, enabling **sub-180ms** query latency over previously unsearchable content.

EDUCATION

University of California, San Diego

Sep 2024 – Mar 2026

Master of Science in Data Science

Indraprastha Institute of Information Technology, Delhi

Sep 2022 – Sep 2023

Post Graduate Diploma in Data Science & Artificial Intelligence

University of Delhi

Jul 2018 – Jun 2021

Bachelor of Management Studies

PROJECTS & RESEARCH

Flare: ML-Powered Anomaly Detection Pipeline | [GitHub](#)

Jan 2026 – June 2026

- Applied Drain3 template mining, Isolation Forest, and DBSCAN to **11.2M log lines** across **575K blocks**, hitting **0.642 F1** held-out at **~40ms** per block, CPU-only.

OneBusAway / maglev: OSS Contributions | [GitHub](#)

Jul 2026 – Present

- Landed **9** upstream fixes to this Go transit REST API from **9** spec-conformance issues I filed.

TRAI: AI Mobile App to Reduce the Gap Between Triage and Care | [IEEE Publication](#)

Jul 2025

- Built the BioBERT decision-support service; **16s** → **5s** inference via caching and batching, RAGAS eval.

TECHNICAL SKILLS

Languages: Python, Go, SQL, Java

ML Systems: PyTorch, torch.compile, torch.library, FlashAttention, Nsight Systems, Diffusers, Quantization

Backend & Infra: FastAPI, REST APIs, Docker, Kubernetes, CI/CD, GitHub Actions, Linux, MCP

Distributed & Data Systems: Kafka, Spark, Airflow, Elasticsearch, MongoDB, Cosmos DB, Azure Queue Storage, Redis

Observability & Cloud: Prometheus, OpenTelemetry, AWS (Lambda, Step Functions, S3), Azure